

AN OPTIMIZED RESIDUAL NETWORK FOR EMOTION RECOGNITION BASED ON A MULTI-FEATURES FUSION TECHNOLOGY AND ELECTROENCEPHALOGRAPHY SIGNALS

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Presently, a significant number of students enrolled in colleges and universities encounter mental health challenges, including academic stress and difficulties in interpersonal relationships. These factors contribute to a decline in mental well-being and necessitate prompt intervention and assistance. The automation of mental health identification in students greatly benefits from the implementation of intelligent emotion recognition systems. Through the analysis of students' emotional states, educational institutions can gain a deeper understanding of students' mental well-being, enabling them to promptly identify and address any issues and offer more complete care and support to students. This paper proposes a method for automated emotion recognition that is driven by data. The method is developed based on the provided background information. The proposed approach incorporates a feature fusion mechanism and an attention mechanism into the ResNet-34 model. This integration enhances the model's capability to analyze intricate details and subsequently improves its classification performance when applied to electroencephalography (EEG) signals. This paper introduces several key innovations. First, an optimization technique is applied to the input component of the model, enabling multifeature fusion. Second, an attention mechanism is incorporated after the residual network module, enabling the model to prioritize parts that contribute to classification and enhance feature extraction. Finally, the network parameters are optimized using both softmax loss and center loss functions. The findings from the analysis of the sentiment EEG public data SJTU Emotion EEG Dataset (SEED) indicate that the proposed sentiment recognition approach not only enhances the

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classification performance of the model on the sentiment EEG data but also improves the stability of the results. This paper presents a novel approach that enables automatic and efficient recognition of students' emotions on commonly used platforms. The findings of this study hold significant implications for mental health assessment and detection in real-life production settings, offering substantial reference value in this domain.

Keywords: Emotion recognition; ResNet-34; EEG signaling; feature fusion; attention.

1. Introduction

With the development of society and the change in education mode, schools undertake the important mission of cultivating future talent, and students' mental health has become a crucial part of the school education system. Automated and intelligent monitoring can collect and analyze students' mental health information in a robust and versatile way with the help of advanced technology and realize timely and precise intervention and support. First, automated and intelligent monitoring can greatly improve the monitoring efficiency and enable schools to have a more comprehensive understanding of students' psychological conditions. Traditional manual investigation and analysis methods are less efficient, while automation technology can realize the rapid processing and analysis of large amounts of data, thereby providing schools with more comprehensive information on students' mental health. Through an automated monitoring system, students' psychological problems can be detected in time, providing schools with a scientific basis for formulating appropriate mental health interventions and avoiding the expansion and deepening of the problems. Second, automated and intelligent monitoring can reduce the workload of teachers and counselors so that they can focus more on mental health support and education. Teachers and counselors play a crucial role in schools, yet manually collecting and analyzing mental health data often consumes a great deal of time and energy. With an automated monitoring system, tedious data processing work can be left to machines, enabling teachers and counselors to devote more energy to face-to-face communication with students and mental health support, improving work efficiency and quality. In addition, automated and intelligent monitoring can improve the relevance and personalization level of intervention and better meet the diverse mental health needs of students. Each student's psychological state and needs may vary, and traditional generic interventions may not be able to fully meet the needs of all students. An automated monitoring system can provide personalized mental health support programs according to the specific situation of each student and the results of data analysis to more accurately help students solve their psychological problems and improve their mental health. In summary, the implementation of automated and intelligent monitoring of students' mental health involves not only the application of modern technology but also the upgrading and improvement of the school education system. By efficiently and accurately collecting and analyzing students' mental health information and providing schools with a scientific basis for implementing personalized interventions, it will help improve the

quality of school education, cultivate healthier and more positive, and make greater contributions to the development of society. Therefore, schools should actively promote the construction and application of automated and intelligent monitoring systems to protect students' mental health.

Emotion recognition technology based on electroencephalography (EEG)¹ has important application prospects for the automated and intelligent monitoring of students' mental health in schools. Through EEG technology, students' EEG activities can be monitored in real time to obtain objective data about their emotional states. There is a close correlation between this emotion recognition technology and monitoring students' mental health. First, it can provide more accurate and objective information about mental health. Compared with traditional questionnaires or face-to-face communication, EEG technology can directly reflect students' brain activity without relying on their subjective answers, thereby reducing the possibility of information distortion and improving the accuracy and reliability of monitoring results. Second, EEG-based emotion recognition technology can realize real-time monitoring and timely intervention. By automating the analysis of students' EEG data, the monitoring system can immediately issue an alert when students' emotions are abnormal or tense, enabling teachers and counselors to take timely measures to provide students with timely psychological support and intervention. In addition, EEG technology also has strong personalized features, which can provide personalized mental health support programs according to the EEG activity patterns of each student and better meet the individualized mental health needs of students. However, care needs to be taken to protect students' privacy and the legal use of this information and to ensure that the monitoring system operates in accordance with regulatory and ethical requirements.

Based on the above analysis, many researchers have carried out EEG-based emotion analysis studies. Goshvarpour *et al.* used EEG to analyze the performance of emotion processing in different brain regions.² The optimal EEG electrode locations were selected by lag Pongalai measurements and the standard low resolution electromagnetic tomography (sLORETA) method. The study evaluated three feature reduction algorithms and found that random subset feature selection (RSFS) performed best. In addition, the study suggested that the choice of EEG electrodes positively affects the classification performance. Pandey *et al.* explored the feasibility of using EEG for emotion recognition, which is particularly applicable in situations such as facial deformities where traditional facial data cannot be accurately acquired.³ The study utilized convolutional neural networks (CNNs) to classify the emotion of wavelet images from EEG data, and the corresponding images were acquired by successive wavelet transforms. The proposed model is subject-independent and aims to extract emotion-related features from EEG data regardless of the data source. Asadzadeh *et al.* applied emotion recognition results to human-computer interface interaction.⁴ The study was conducted by employing the standardized sLORETA method, where we successfully mapped scalp sensors to brain sources, thereby clearly defining the pattern of each emotion. Next, a graph neural

network (GNN) was employed for EEG-based emotion recognition, representing the sLORETA sources as nodes of the underlying graph. Finally, a GNN classifier was used to identify unknown emotion labels. In addition to EEG-based emotion recognition, there is also interest in emotion recognition based on data such as audio,^{5,6} video,^{7,8} and facial expressions.^{9–11} Compared to the recognition of data such as audio, video or face data, EEG-based emotion recognition is able to provide information that directly reflects brain activity, which enables a more objective and accurate way of recognizing emotions. This means that it is able to avoid being affected by external environmental noise or surface emotional expression, providing a more reliable data source for emotion recognition. At the same time, EEG-based emotion recognition can also overcome the limitations of language and cultural differences, making it more widely applicable, thereby facilitating cross-cultural and cross-linguistic emotion research. Therefore, this paper proposes an EEG data-driven emotion recognition method, which introduces a feature fusion mechanism and an attention mechanism on the basis of the ResNet-34 model, thereby enhancing the model's ability to process details and improving the model's classification performance on EEG signals.

2. Research Basis

2.1. Introduction to EEG

EEG is a technique for monitoring neural activity in the brain. By attaching tiny electrodes to the scalp, it can capture the faint electrical signals emitted by neurons in the brain. Graphs are generated from the amplified signals. The electrical activity of neurons in the brain is the primary source of EEG data. Electrodes placed on the scalp can capture the modest electric current produced when neurons in the brain are activated. Among the many features of EEG, EEG records brain activity in real time, enabling the visualization of immediate changes in the brain. Compared to other methods, EEG's accuracy in pinpointing brain activity in specific regions is lacking; therefore, the resulting picture is murky at best. However, EEG does not involve any risky or painful procedures, only attaching electrodes to the scalp. In conclusion, EEG is a crucial tool for studying brain function. Neuroscience, clinical medicine, and psychological experiments all make extensive use of it.

Different emotions result in distinct electrical impulses being sent from the brain. Similar to "electrocardiography" except with the recording of brain activity, EEG can assist us in recording the electrical signals that occur in the brain when we are in different emotional states. When a person is feeling particularly elated, for instance, a certain set of electrical signals is generated in the brain. In contrast, a different set of electrical signals is generated in the brain when a person feels anxious or sad. Scientists can deduce the current emotional state from the unique properties of these electrical signals. An electroencephalogram can reveal a person's emotional state in the same way as facial expressions can. This has the potential to be highly useful in the study of emotions and the analysis of brain activity associated with various states of mind. A physician can use EEG to track brain activity and determine

whether a treatment is having the desired effect on a patient's mental state, for instance. In general, EEG serves as a useful "mood detector" by shedding light on the peculiarities of the brain's operation while a person is feeling particularly happy or sad.

2.2. Theories of emotion

Feelings, bodily changes, and the subsequent actions people take all contribute to what we call an emotional state. Emotions are transient mental states that can be communicated verbally, behaviorally, physiologically, and neurologically in response to a variety of environmental cues. Emotion, as used in affective neurology, is distinguished from mood or feeling, which emphasizes the individual's subjective experience of emotion; the state of mind lasts for an extended period of time. It is a vast category of feelings that span from barely perceptible to highly noticeable. Emotion, on the other hand, encompasses not only feelings but also mental states. There are two main approaches to emotion representation. The first theory contends that natural selection is responsible for the development of core feelings. Emotions, in this view, are discrete entities that can be represented along a single axis with values ranging from tiny to large to indicate different sorts of emotions. One-dimensional discrete emotion models typically consist of six emotions: anger, disgust, fear, happiness, sorrow, and surprise.¹² Psychologists frequently utilize models based on eight primary emotions¹³: anger, fear, sadness, disgust, surprise, curiosity, acceptance, and happiness. The emotions contained in the one-dimensional emotion model are countable, but they are also the most fundamental; many additional emotions that are not included in the model can be constructed from them.

The cognitive process viewpoint, on the other hand, considers emotional experiences to be processes that must be completed. Emotional change is therefore given greater weight in cognitively oriented models of emotion. These feelings can be assigned to various locations in the universe, depending on their nature. The arousal-valence (AV) model is popular because it can be represented in a two-dimensional space.¹⁴ In this framework, arousal refers to the level of arousal (from relaxed to stimulated), and valence represents the level of enjoyment (from unpleasant to enjoyable). Different discrete affective states can be expressed by decomposing and mapping them in different regions, as shown in the AV model, where the horizontal axes represent the intensity of pleasure and the vertical axes represent the intensity of arousal, and where values from small to large indicate that the state is from weak to strong.

The PAD theory of emotion¹⁵ organizes emotions into a three-dimensional space, where P represents pleasure and highlights the different degrees of variation in the degree of pleasure, A represents activation and reflects the variation in the degree of arousal of an individual, and D represents dominance and reflects the degree to which an individual is in control of the situation and others. Finally, as indicated in Table 1, we can express various emotional states through the values of these three dimensions.

Table 1. Classification results of three-dimensional emotion combinations.

+P+D+A-	Willing	-P-A-D-	Nonsensical
+P+A-D-	Dependencies	-P-A+D-	Disdain
+P-A+D-	Unwind	-P+A-D-	Apprehensive
+P-A+D-	Meek	-P+A+D-	Animosity

3. Emotional Recognition Methods Based on Improved ResNet Models

3.1. Overall framework for emotion recognition

Among the most widely used ResNet models¹⁶ are ResNet-34, ResNet-50, ResNet-101, and ResNet-152. ResNet-34 stands out as a lightweight option among these types, making it ideal for use in this investigation. When applied to the EEG classification task, however, the ResNet-34-based deep residual network has two glaring weaknesses. First, in regard to data feature extraction, different types of EEG data might readily be mistaken for one another because they share so many commonalities. Second, in terms of classification, it can be challenging for the commonly used Softmax loss function to effectively categorize two separate EEG data because they may be similar in various ways. To increase recognition accuracy, this research suggests an optimized residual network structure, which is a model capable of accurately focusing on considerably distinct regions in EEG. Input, convolutional, pooling, fully connected, and output layers make up the five-part architecture of a deep residual network. Figure 1 presents a representation of the sentiment recognition framework of this paper. The total model consists of three steps: (1) pre-processing the EEG data to be utilized in training; (2) using the optimized ResNet-34 network as a feature extractor; and (3) using an optimized loss function for image classification and output.

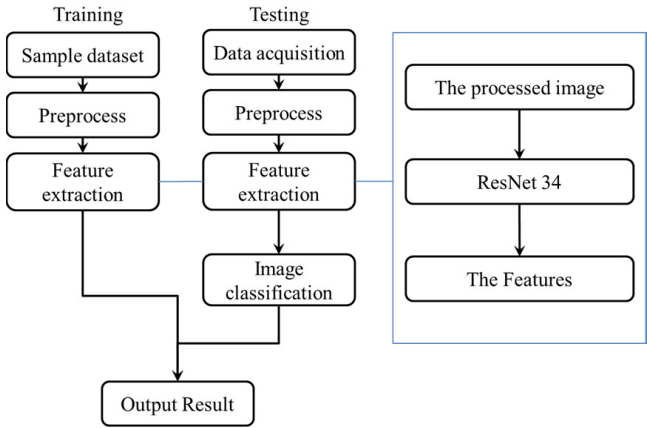


Fig. 1. Emotion recognition framework diagram.

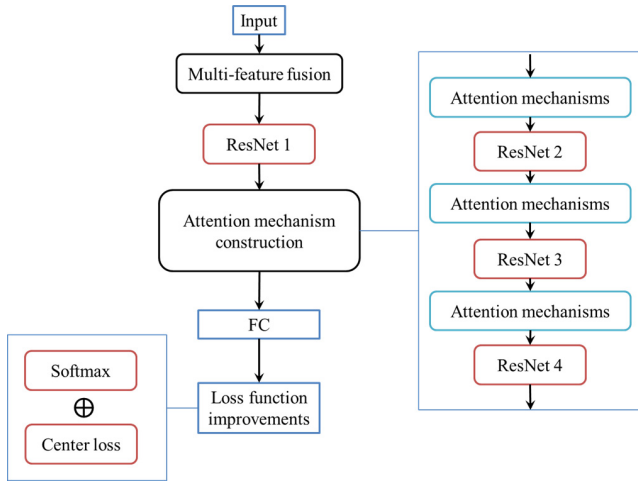


Fig. 2. Optimized deep residual network.

Figure 2 depicts the structure of the optimized residual network model, which has undergone three major improvements. First, the concept of fusing several aspects of the input image was introduced, which improves the network’s ability to grasp the full context of the image. Second, an attention mechanism was added after the residual network module to help the model extract characteristics that are useful for classification. This helps the model perform better. Finally, the network’s parameters are optimized using both the softmax loss and center loss functions, which results in more precise and efficient training. These enhancements enable us to obtain a more robust and precise model with which to address the issue at hand.

3.2. Feature fusion mechanisms

When many feature sources are used, model performance is enhanced by the feature fusion approach. The model can more accurately interpret and forecast complicated data relationships by successfully integrating information from multiple features into a single, more complete input signal. As shown in Fig. 3, this work optimizes the input portion, employs multiple parallel branches to extract features, and then fuses the derived features at the end of each branch to provide multifeature fusion. Three separate paths are implemented, each with a different-sized convolution kernel, to achieve this goal. The convolution operation in the first branch employs a kernel of size 1×1 , whereas the two convolution operations in the second branch employ kernels of sizes of 1×1 and 3×3 . The third fork performs triple convolution with 1-, 3-, and 3-pixel kernels. In this scenario, a 1×1 convolution kernel is used to reduce the network’s overall depth and improve its feature extraction performance. To facilitate the extraction of features of varying sizes, the network’s subsequent layer makes use of a 3×3 convolution operation between the second and third branches. The step size of the convolution process in all three branches is fixed at 1 to maintain

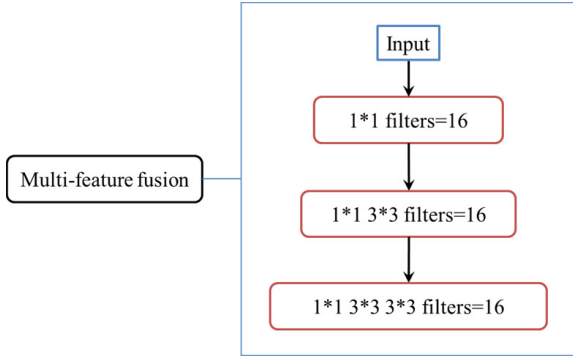


Fig. 3. Multifeature fusion mechanism.

the original input image size. At the end of the two forks, the features are fused together to boost both the model's feature extraction and nonlinear expression capabilities.

The addition of a channel attention mechanism is the main benefit of ECA-Net. However, its computational complexity is modest compared to those of other attention models. This is because channel attention is computed using elementary procedures, such as pooling over a global average and having all layers completely connected. This improves the performance of ECA-Net in environments with restricted computing resources.

Multifeature fusion operates according to the following principle: Following feature extraction from three distinct dimensions of EEG data, the following feature vectors are expected to be present in the data:

$$\begin{aligned}
 &F_{1,1,1}, F_{2,1,1}, \dots, F_{16,1,1}; \\
 &F_{1,1,2}, F_{2,1,2}, \dots, F_{16,1,2}; \\
 &\vdots \\
 &F_{1,1,15}, F_{2,1,15}, \dots, F_{16,1,15}; \\
 &F_{1,1,16}, F_{2,1,16}, \dots, F_{16,1,16}.
 \end{aligned} \tag{1}$$

We let $F_{a,b,c}$ denote the EEG blocks of size a , c belonging to the b th category. Next, the Euclidean distance is computed between the extracted features from each branch and the standard samples. Subsequently, the membership degree is determined to derive the membership matrix.

$$\begin{aligned}
 &u_{1,1,1}, u_{2,1,1}, \dots, u_{16,1,1}; \\
 &u_{1,1,2}, u_{2,1,2}, \dots, u_{16,1,2}; \\
 &\vdots \\
 &u_{1,1,15}, u_{2,1,15}, \dots, u_{16,1,15}; \\
 &u_{1,1,16}, u_{2,1,16}, \dots, u_{16,1,16}.
 \end{aligned} \tag{2}$$

The level of membership is determined by the following criteria:

$$u_{a,b,c} = \frac{1}{1 + \left(\frac{d_{a,b,c}}{d_{b,c}}\right)^\varepsilon}, \quad (3)$$

$$d_{a,b,c} = \|F_{b,c} - F_{a,b,c}\|, \quad (4)$$

$$d_{b,c} = \frac{1}{10} \sum_{i=1}^{10} d_{a,b,c}. \quad (5)$$

In this study, the overall unclassified EEG is denoted by $F_{b,c}$, while the standard image feature vector is represented by $F_{a,b,c}$. The Euclidean distance between these two vectors is denoted by $d_{a,b,c}$. Additionally, the fuzzy factor $F_{b,c}$ is utilized to regulate the degree of fuzziness, and it assumes values within the range of 0 to 1. Through experimental analysis, it has been determined that the optimal classification performance is achieved when the value of ε is set to 0.2. Consequently, the degree of affiliation is determined by the value of $\varepsilon = 0.2$, which is utilized in the computation of the affiliation matrix.

The fusion results are generated by aggregating the affiliation matrices in the column direction for each EEG category. This process yields the affiliation vector $[u_1, u_2, u_3, u_4]^T$, which represents the final feature fusion result.

3.3. Attention mechanisms

The attention mechanism of a neural network is commonly understood as the capacity of the network to selectively prioritize certain feature vectors by assigning distinct weights to them. We let $x \in R^d$ represent the input vector, $z \in R^k$ denote the feature mapping, $\alpha \in [0, 1]^k$ represent the attention vector, $g \in R^d$ represent the attention segment, and $f_\Phi(x)$ represent the attention network, which can be a distinct branch within the same network with varying parameters. The attention vector and segment can be expressed in the following manner:

$$\alpha = f_\Phi(x), \quad (6)$$

$$g = \alpha \cdot z. \quad (7)$$

In the context of element-by-element multiplication, denoted by $\cdot$, the attention weight α is constrained to the range of values between 0 and 1, namely, $\alpha \in [0, 1]^k$. When the value of α is either 0 or 1, the attention is referred to as hard attention. Soft attention refers to the attentional weight being constrained within the range of 0 to 1. This particular aspect is of relevance and will be addressed in the present work.

The research presents a developed attention mechanism, as depicted in Fig. 4, that enables the model to choose to focus its attention on specific portions of an image. This attention mechanism enhances the classification function of the model. The surface characteristics of various categories of waste in daily life

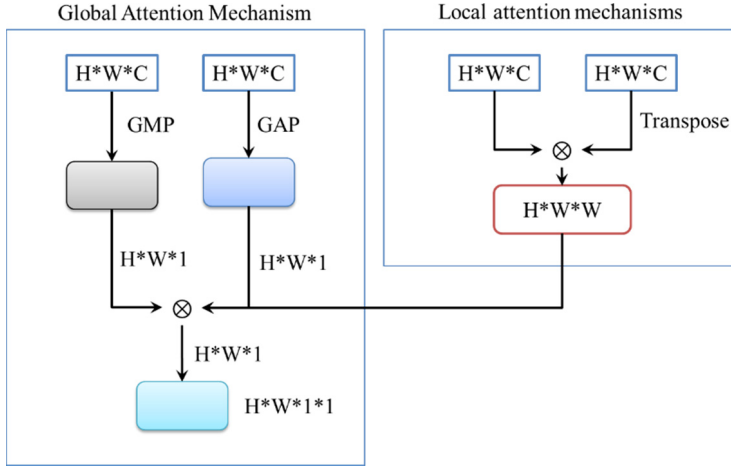


Fig. 4. Schematic diagram of the attention mechanism.

exhibit limited variation, which hinders the effective classification of waste images. Therefore, this paper aims to investigate methods for enabling the model to accurately identify and focus on the regions of significant differences within images. This concept consists primarily of two distinct components, namely, the global attention system and the local attention mechanism.

As depicted in Fig. 4, the global attention mechanism pertains to the phenomenon wherein the neurons within the initial residual network module acquire the characteristics of a limited area within the image. This region, known as the neuronal receptive region, has a size of 16×16 , which is determined by the dimensions of the filters employed. In the subsequent residual network module, a convolution operation is executed on a 16×16 region, similar to the preceding layer. Consequently, the input image necessitates a substantial receptive region. As the number of network layers increases, each neuron is capable of representing an increasingly extensive receptive region.

A localized attention mechanism refers to the utilization of a Gram matrix, designated O_L , which is obtained by multiplying the matrix O_i with its transpose O_i^T . This attention mechanism module is produced within each residual network structure, resulting in an output O of size $H \times W \times C$. The module itself has a size of $H \times W \times C$.

$$O_L = O_i \times O_i^T. \quad (8)$$

The aforementioned procedure involves the multiplication of pixel values inside the feature map by their respective matrices. This results in an amplification of pixel values that were initially larger, while simultaneously diminishing the values of pixels that were originally smaller. By employing this method, regions of the image that necessitate particular scrutiny are magnified, hence enhancing the precision of categorization decisions and eliminating extraneous feature data.

Finally, O_A , O_M , and O_L , which have been subjected to different pooling operations, are subjected to a multiplication operation, i.e., one extracted global feature O is multiplied by two local features to obtain the overall features of the image.

$$O = O_L \times O_A \times O_M. \quad (9)$$

4. Experimental Results and Analysis

4.1. Experiment-related settings

Throughout the experimental portion, the SEED dataset was utilized. The SEED dataset consists of EEG signals evoked in 15 healthy people (seven men and eight females, with a mean age of 23.27 years) by having them watch 15 Chinese movie clips. The Eysenck Personality Inventory and Philip's Questionnaire were used to ensure that the subjects could accurately express the emotions that were depicted in the 15 movie clips. The films included positive, neutral, and negative feelings. The dataset was segmented into a training set, a validation set, and a test set using the proportion 5:3:2 as the dividing factor. The model was trained with the parameters presented in Table 2.

The traditional classification accuracy (Accuracy) and recall (F1) were chosen as the evaluation metrics. For the emotion EEG recognition experiments, an Intel Core i7-10700K processor with eight cores, sixteen threads, and a base frequency of 3.8 GHz was used in this paper. In addition, 32 GB DDR4 RAM was equipped to support large-scale model training and data processing. To improve the data reading and writing speed, an SSD (solid-state drive) was used as the storage device in this study. To accelerate deep learning model training, we also used an NVIDIA GeForce RTX 3080 graphics card. The above describes the hardware environment. The software environment of this study was as follows: The operating system was Ubuntu, and the programming language was Python 3.0. In addition, deep learning frameworks such as TensorFlow and PyTorch were used to implement the ResNet-34 model.

Table 2. Details of parameter settings for the model.

Parameter name	Setup details
Activation function	ReLU
Batch size	64
Loss function	Center loss and softmax loss
Pooling	Average pooling
Dropout	0.5
Optimizer	Adam
Epochs	60
Learning rate	0.01

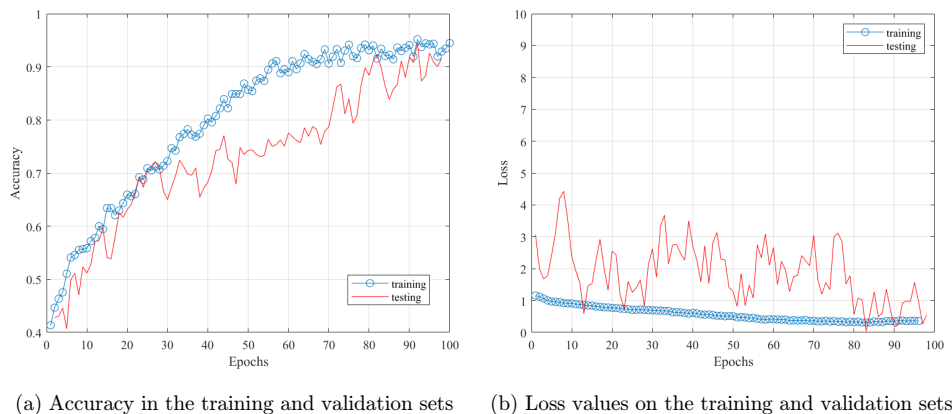


Fig. 5. Emotion recognition results based on each model.

4.2. Experiments to analyze the performance of this model

To evaluate the recognition performance of this paper's model for emotion EEG, the training set and validation set were first fed into the proposed model, and the obtained emotion recognition performance results are shown in Fig. 5.

First, Fig. 5(a) shows that the model proposed in this paper achieves a high accuracy on the training set, which may indicate that the model is able to fit the training data well but does not necessarily perform well on unseen data. Therefore, we take a closer look at the accuracy rates obtained on the validation set. The accuracy on the validation set is overall lower than that on the training set, but the difference is not very large. It can also be observed that the accuracy curve obtained on the validation set has some volatility, which indicates that the model in this paper performs worse on the unseen dataset than on the training set. In Fig. 5(b), it can be observed that the loss values on the training set are low, which indicates that the model of this paper performs better on the training set. The loss value curves on the validation set are slightly more volatile, and the overall loss values are all larger than those on the training set. This indicates that there are some differences between the model's predictions on the validation set and the actual values, but the differences are not large; hence, overall, the gap between the loss values on the training and validation sets is small. In summary, the model in this paper is more effective in recognizing emotional EEG data.

4.3. Performance comparison experiments with other similar models

The model in this paper aims to improve the model's classification performance on EEG as much as possible without increasing the computational cost. To validate the effectiveness of the models used, this paper compares the classification performance of several similar models on the publicly available dataset SEED. The compared models are as follows: ResNet-18¹⁷: a lighter version of the ResNet family that

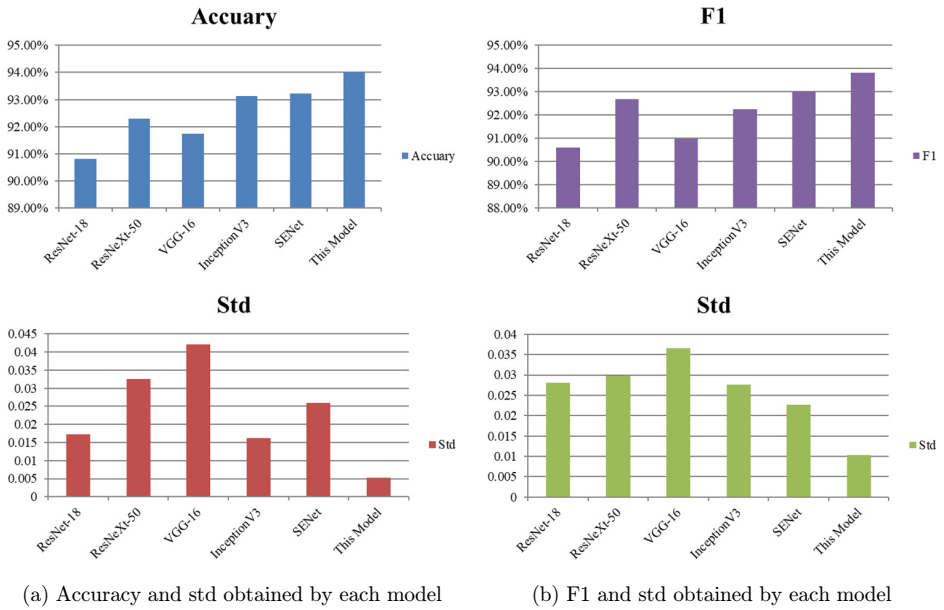


Fig. 6. Comparison of emotion recognition results for each model.

contains 18 convolutional layers; ResNeXt-50¹⁸: ResNeXt is an extension of ResNet, and ResNeXt-50 has 50 layers and introduces grouped convolution within the convolutional layers; VGG-16¹⁹: VGG is a classic CNN architecture, and VGG-16 contains 16 convolutional layers; InceptionV3²⁰: InceptionV3 is a version of the Inception family that performs well on image classification tasks; and SENet²¹: SENet introduces an “attention mechanism” to enable the network to automatically focus on features that are important for classification. The results of each model on the SEED dataset are shown in Table 3. To visualize the performance differences among the models, Fig. 6 plots the experimental results shown in Table 3.

(1) Among all the models, the model in this paper has the highest accuracy rate of 94.02%, outperforming the other models. The accuracy rates of the ResNeXt-50 and SENet models are also quite high, reaching 92.30% and 93.23%, respectively. They slightly outperform the ResNet-18, VGG-16 and InceptionV3 models in terms of accuracy. The accuracy rates of the ResNet-18 and VGG-16 models are 90.82% and 91.73%, respectively, which are slightly lower than those of the other models.

Table 3. Sentiment recognition results for each model.

Index\model	ResNet-18	ResNeXt-50	VGG-16	InceptionV3	SENet	This model
Accuracy	90.82%	92.30%	91.73%	93.12%	93.23%	94.02%
Std	0.0172	0.0325	0.0421	0.0163	0.0259	0.0052
F1	90.59%	92.67%	90.98%	92.25%	93.03%	93.81%
Std	0.0281	0.0298	0.0365	0.0276	0.0227	0.0103

(2) The model in this paper also has the highest F1 score of 93.81%. This shows that the model performs well in considering both accuracy and recall. The ResNeXt-50 and SENet models also perform well in terms of F1 scores, with values of 92.67% and 93.03%, respectively. The F1 scores of the other models are slightly lower than those of the current model but still remain above 90%, showing good performance.

(3) Standard deviation is a measure of the stability of the experimental results. A smaller standard deviation indicates more stable experimental results. In terms of the standard deviations of both accuracy and F1 score, the current model has the most stable performance, with values of 0.0052 and 0.0103, respectively. The standard deviations of the other models are also within a reasonable range, suggesting that their performance remains relatively stable over multiple experiments.

In summary, according to the data provided, the current model performs well in terms of accuracy and F1 score and has high stability. On the other hand, we can also see that the improvement in the recognition performance of the model proposed in this paper for emotional EEG is not obvious, but the stability of this model is indeed significantly better than those of the other models. Therefore, this model can be considered for application in real-life scenarios.

5. Conclusions

In this study, we proposed an emotion recognition method that integrates an optimized ResNet-34 model with EEG signals, aiming for application to student mental health monitoring. By introducing a feature fusion mechanism and an attention mechanism, we optimized the ResNet-34 model, strengthening the model's ability to process details and improving the classification performance on EEG signals. Our research achieved significant innovations in the following three aspects. First, we optimized the input part of the model to achieve multifeature fusion. This not only improves the feature extraction capability of the model but also provides richer information inputs to the model, which enables the model to comprehensively understand the emotion information in EEG signals. Second, we introduced an attention mechanism after the residual network module, which enables the model to focus on the parts that have important contributions to emotion categorization. This initiative effectively improves the feature extraction function of the model, which enables the model to capture the emotional features in the EEG signals more accurately. Finally, we adopted a strategy of using both softmax loss and center loss functions for network parameter optimization. This combined optimization method enables the model to maintain sensitivity to emotion classification during the training process and ensures that the feature representation of the model has a certain degree of compactness, thereby making the model more generalizable. Experiments demonstrated that the proposed emotion recognition method achieved significant results on the SEED dataset. The method not only improves the model's classification performance on emotion EEG data but also increases the stability of the results, providing an effective and reliable technical tool for monitoring students'

mental health. However, this study had various limitations. For example, our experimental data were mainly based on the SEED dataset, which may have some data bias. In the future, we plan to expand the experimental data sources and increase the sample diversity to improve the generalization ability of the model. We believe that these efforts will further promote the development and application of emotion recognition technology in education.

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